
***Environmental Intelligence and Sustainability Analytics Platform
Machine Learning–Driven Environmental Risk Assessment, Forecasting, &
Decision Support System***

Team/Group 14

Week 5 – Data Engineering, Analytics and Forecasting

Executive Summary

The **Environmental Intelligence and Sustainability Analytics Platform** was developed to analyze environmental conditions, assess sustainability performance, forecast future climate trends, and support data-driven environmental decision-making. The solution integrates three key environmental domains—**climate change, air quality, and forest sustainability**—into a unified analytical framework that enables comprehensive environmental assessment and forecasting. The project combines historical global temperature anomaly data (1880–2026), air quality monitoring data comprising more than **29,000 records**, and forest cover datasets to generate meaningful environmental intelligence indicators. Through extensive data engineering and feature engineering processes, several advanced metrics were developed, including the **Climate Severity Index (CSI)**, **Pollution Exposure Index (PEI)**, **Forest Sustainability Index (FSI)**, **Environmental Health Index (EHI)**, **Environmental Risk Score**, and **Sustainability Score**.

1. Introduction

Environmental sustainability has become one of the most critical challenges of the twenty-first century. Climate change, air pollution, and deforestation have significant impacts on both ecological systems and human well-being.

Organizations and policymakers require data-driven approaches to:

- Monitor environmental performance
- Assess sustainability risks
- Forecast future environmental trends
- Support strategic decision-making

This project addresses these needs through an integrated environmental intelligence platform combining data engineering, machine learning, forecasting, and business intelligence techniques.

2. Project Objectives

The primary objectives of this project are:

Objective 1: Integrate multiple environmental data sources.

Objective 2: Perform feature engineering to generate environmental intelligence indicators.

Objective 3: Develop machine learning models to forecast future environmental trends.

Objective 4: Evaluate model performance using statistical metrics.

Objective 5: Conduct scenario analysis for future environmental outcomes.

Objective 6: Provide Power BI-ready datasets for business intelligence visualization.

3. Datasets Used

3.1 Temperature Dataset: The temperature dataset contains annual global temperature

anomalies from 1880 to 2026.

Key fields:

Year

Jan-Dec

J-D (Annual Mean Anomaly)

Purpose:

- Climate trend analysis
- Forecasting
- Climate Severity Index calculation

3.2 Air Quality Dataset

Contains: 29,531 records

Variables:

AQI

PM2.5

PM10

NO

NO2

NOx

NH3

CO

SO2

O3

Benzene

Toluene

Xylene

Purpose:

- Pollution Exposure Index
- Air Quality Analytics
- Risk Assessment

3.3 Forest Cover Dataset

Contains forest cover estimates between 2005 and 2017.

Purpose:

- Forest Sustainability Index
- Environmental Health Index
- Deforestation Monitoring

4. Data Engineering and Preprocessing

The datasets underwent extensive preprocessing.

Activities included:

Cleaning

- Missing value handling
- Data type correction
- Column standardization

Transformation

- Time-based feature generation
- Trend extraction
- Forest interpolation

Validation

- Data quality checks
- Missing value reporting
- Consistency verification

5. Feature Engineering

Several analytical indicators were developed.

5.1 Climate Severity Index (CSI)

Measures relative climate stress using temperature anomalies.

Formula:

$$CSI = (\text{Temperature} - \text{Min Temp}) / (\text{Max Temp} - \text{Min Temp}) \times 100$$

5.2 Pollution Exposure Index (PEI)

Measures pollution impact using air quality indicators.

Formula:

$$PEI = 0.5 \times PM_{2.5} + 0.3 \times PM_{10} + 0.2 \times AQI$$

5.3 Forest Sustainability Index (FSI)

Measures relative forest sustainability.

Formula:

$$FSI = \text{Forest Cover} / \text{Maximum Forest Cover} \times 100$$

6. Environmental Intelligence Layer

6.1 Environmental Health Index (EHI)

Formula:

$$EHI = 0.4 \times \text{Forest Score} + 0.3 \times \text{Climate Score} + 0.3 \times \text{Pollution Score}$$

Result: Environmental Health Index = 60.18

Interpretation: The environment exhibits moderate overall health with opportunities for improvement through pollution reduction and conservation initiatives.

6.2 Environmental Risk Score

Result: Risk Category = Moderate

Interpretation:

Environmental risks exist but are manageable through policy intervention and sustainability programs.

6.3 Sustainability Score

Result: Sustainability Category = Good

Interpretation:

Current environmental conditions indicate generally sustainable performance with room for improvement.

7. Machine Learning Models

Two machine learning models were implemented.

7.1 Linear Regression

Purpose:

- Benchmark model
- Trend forecasting

7.2 Random Forest Regressor

Purpose:

- Non-linear pattern discovery
- High-accuracy forecasting

Parameters:

n_estimators = 300

random_state = 42

8. Model Evaluation

The following metrics were used:

Mean Absolute Error (MAE) - Measures average forecasting error.

Root Mean Squared Error (RMSE) - Measures squared error magnitude.

R-Squared (R^2) - Measures explanatory power.

Best Model - Random Forest Regressor

Results

MAE = 0.028886

RMSE = 0.034721

R^2 = 0.992454

Interpretation:

The Random Forest model achieved excellent predictive performance and was selected as the final forecasting model.

9. Forecasting Analysis

Future temperature anomalies were forecast for:

2027

2028

2029

2030

2031

The forecasting module included:

- Point forecasts
- Confidence intervals
- Forecast uncertainty analysis

Results indicate continued elevated temperature anomalies in future years.

10. Scenario Analysis

Four environmental scenarios were developed.

Baseline - Current environmental trajectory.

Green Policy - Reduced pollution and improved forest cover.

Aggressive Green - Strong sustainability initiatives.

Pollution Growth - Increased pollution levels.

Key Finding

Aggressive Green scenarios resulted in lower projected environmental impact, while Pollution Growth scenarios produced the highest projected temperature outcomes.

11. Clustering Analysis

KMeans clustering was implemented to identify environmental segments.

Number of Clusters: 3

The clustering process grouped environmental conditions into:

Low Risk

Moderate Risk

High Risk

This segmentation supports targeted environmental interventions.

12. Key Findings

The analysis produced several significant insights:

Environmental Health Index (EHI): 60.18

- Indicates a moderate level of overall environmental health with opportunities for improvement through targeted sustainability initiatives.

Environmental Risk Category: Moderate

- Environmental pressures are present but remain manageable through proactive policy intervention and environmental monitoring.

Sustainability Category: Good

- Current environmental conditions demonstrate generally sustainable performance despite increasing climate and pollution-related challenges.

Best Predictive Model: Random Forest Regression

- Achieved exceptional forecasting performance:
- **R² Score:** 0.992454
- **MAE:** 0.028886
- **RMSE:** 0.034721

Forecast Horizon: 2027–2031

- Forecasting results indicate the continuation of elevated temperature anomaly levels, highlighting the importance of long-term climate mitigation strategies.

Scenario Analysis Results

- The **Aggressive Green** scenario produced the most favorable environmental outlook.
- The **Pollution Growth** scenario resulted in the highest projected environmental impact and risk.

Environmental Segmentation

- K-Means clustering identified three distinct environmental profiles, enabling the classification of environmental conditions into actionable risk groups.

13. Business Recommendations

Based on analytical findings:

Recommendation 1 - Implement pollution-control initiatives to reduce AQI and PM2.5 levels.

Recommendation 2 - Increase afforestation programs to improve environmental sustainability.

Recommendation 3 - Strengthen climate adaptation and resilience planning.

Recommendation 4 - Enhance environmental monitoring and data-driven decision-making.

14. Power BI Readiness

The project generates Power BI-ready datasets including:

fact_environment.csv

dim_time.csv

dim_risk.csv

dim_scenario.csv

dashboard_master_dataset.csv

These datasets support interactive dashboards and executive reporting.

15. Conclusion

This project successfully demonstrates the integration of environmental analytics, machine learning, forecasting, clustering, and business intelligence techniques to address sustainability challenges.

Key achievements include:

- Multi-source environmental data integration
- Advanced feature engineering

- Environmental Health Index development
- Machine learning forecasting
- Scenario analysis
- Environmental clustering
- Executive reporting
- Power BI data model preparation

The Random Forest model achieved excellent predictive accuracy with an R^2 score of 0.992454, demonstrating the effectiveness of the developed analytics framework.

The resulting Environmental Intelligence Platform provides a scalable foundation for environmental monitoring, sustainability assessment, and strategic environmental decision-making.

16.Business Impact

The results generated through this project provide significant value for environmental planning and sustainability management:

Strategic Decision-Making

The Environmental Health Index and Risk Score provide management and policymakers with an easy-to-understand view of overall environmental performance, enabling more informed strategic decisions and prioritization of sustainability initiatives.

Risk Identification and Mitigation

The Environmental Risk Engine helps identify key environmental pressure points, allowing organizations to proactively implement mitigation strategies before risks escalate into larger environmental or operational issues.

Forecast-Based Planning

The forecasting framework enables decision-makers to anticipate future temperature trends and environmental conditions rather than relying solely on historical observations. This supports long-term planning, climate resilience initiatives, and sustainability roadmaps.

Scenario-Based Policy Evaluation

The scenario forecasting module allows stakeholders to evaluate the potential impact of various environmental policies before implementation. By comparing Baseline, Green Policy, Aggressive Green, and Pollution Growth scenarios, decision-makers can assess potential outcomes and invest in the most effective sustainability strategies.

Sustainability Monitoring

The Sustainability Score and Forest Sustainability Index provide ongoing mechanisms for tracking environmental performance and measuring progress toward sustainability objectives.

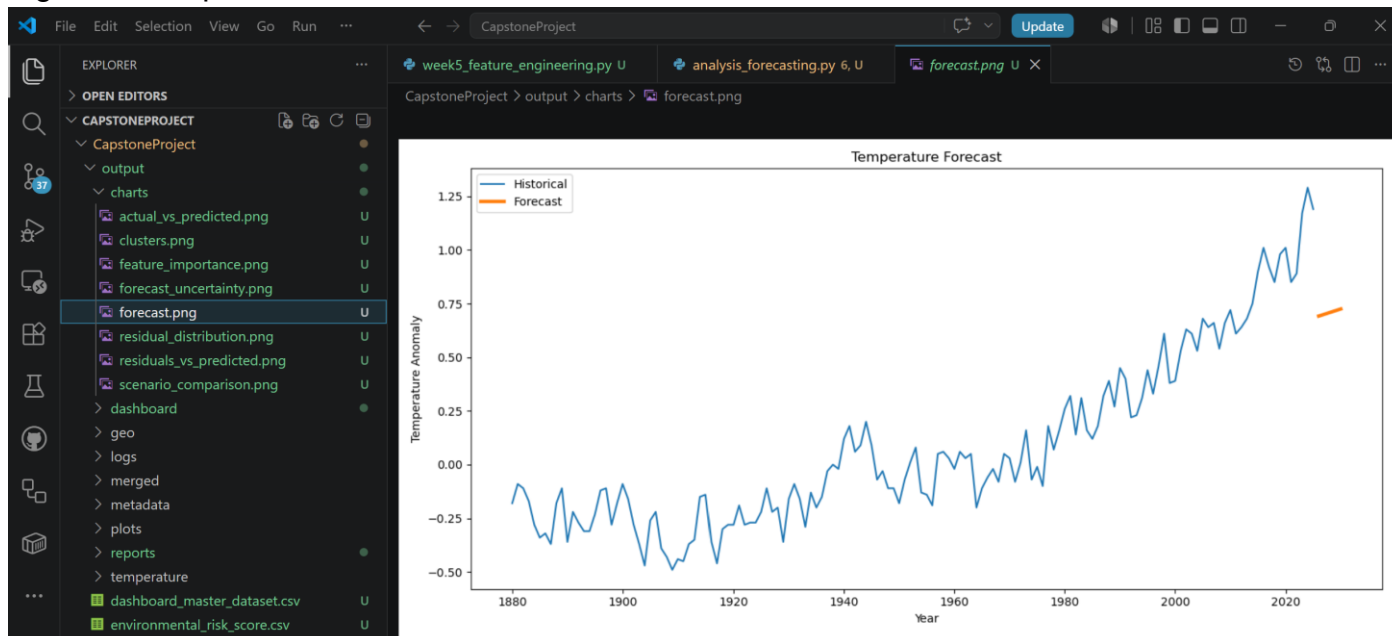
Business Intelligence and Reporting

The project generates Power BI-ready datasets, executive reports, business recommendations, and star-schema data models that support real-time dashboarding, executive reporting, and

organizational sustainability governance.

17. Screenshots and Visual Evidence

Figure 1: Temperature Forecast



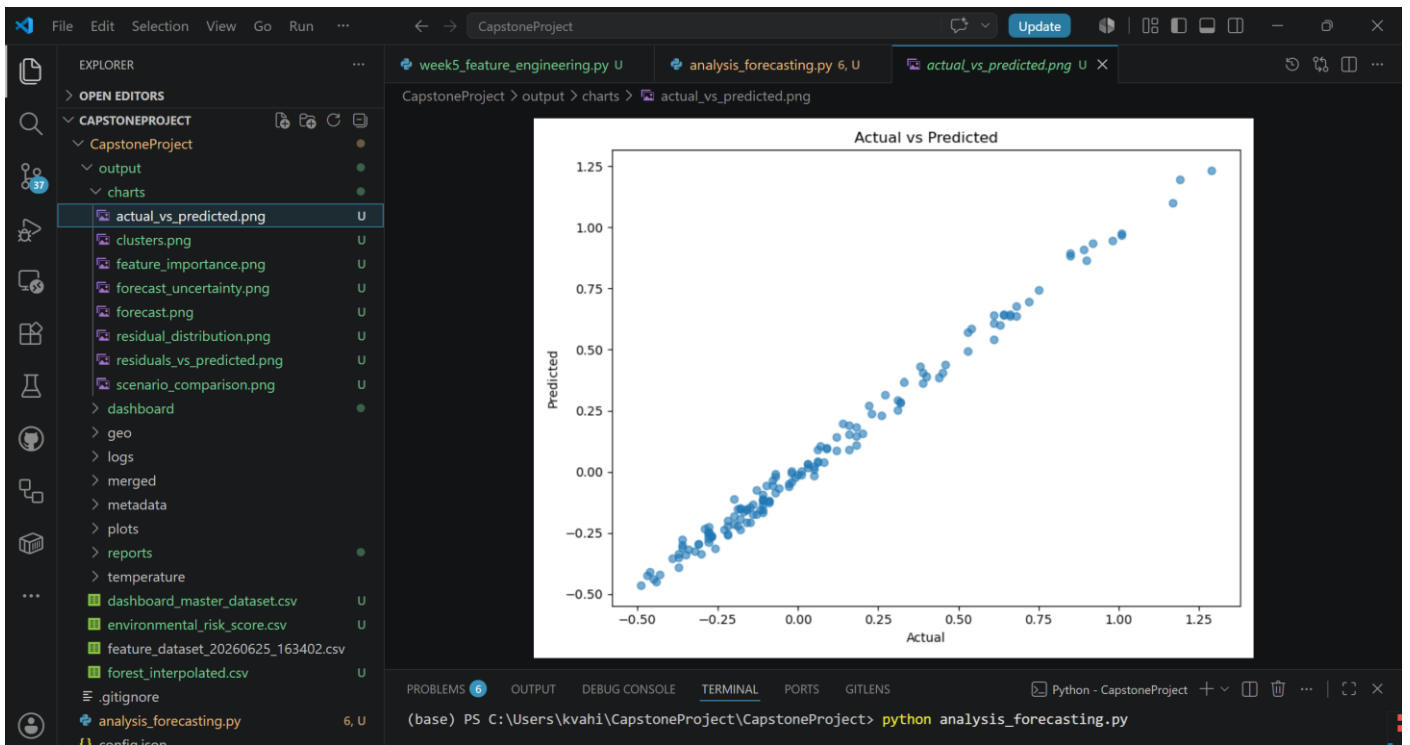
This chart shows historical temperature anomalies from 1880–2026 and forecasted temperature anomalies for the period 2027–2031. The historical data indicates a long-term warming trend, while the forecasting model predicts continued elevated temperature anomalies in future years. This supports the conclusion that climate change impacts are expected to persist without significant intervention.

Figure 2: Forecast Uncertainty Analysis



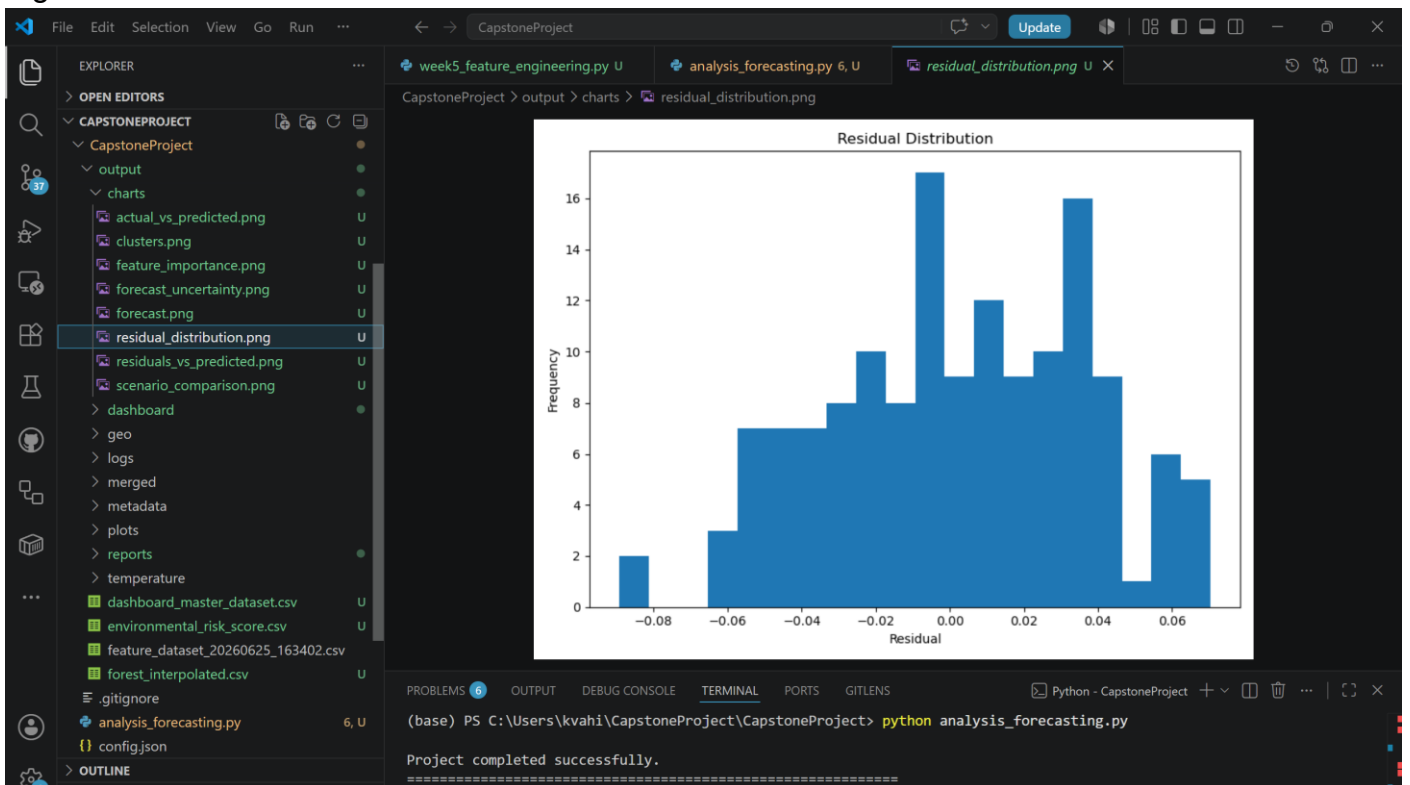
This visualization presents forecast confidence intervals around the predicted temperature anomalies. The shaded region represents the range of possible future outcomes, helping stakeholders understand forecast uncertainty and potential environmental variability.

Figure 3: Actual vs Predicted Values



This chart compares actual temperature anomaly values against model predictions. The close alignment of points demonstrates the strong predictive performance of the Random Forest model and validates its suitability for environmental forecasting.

Figure 4: Residual Distribution



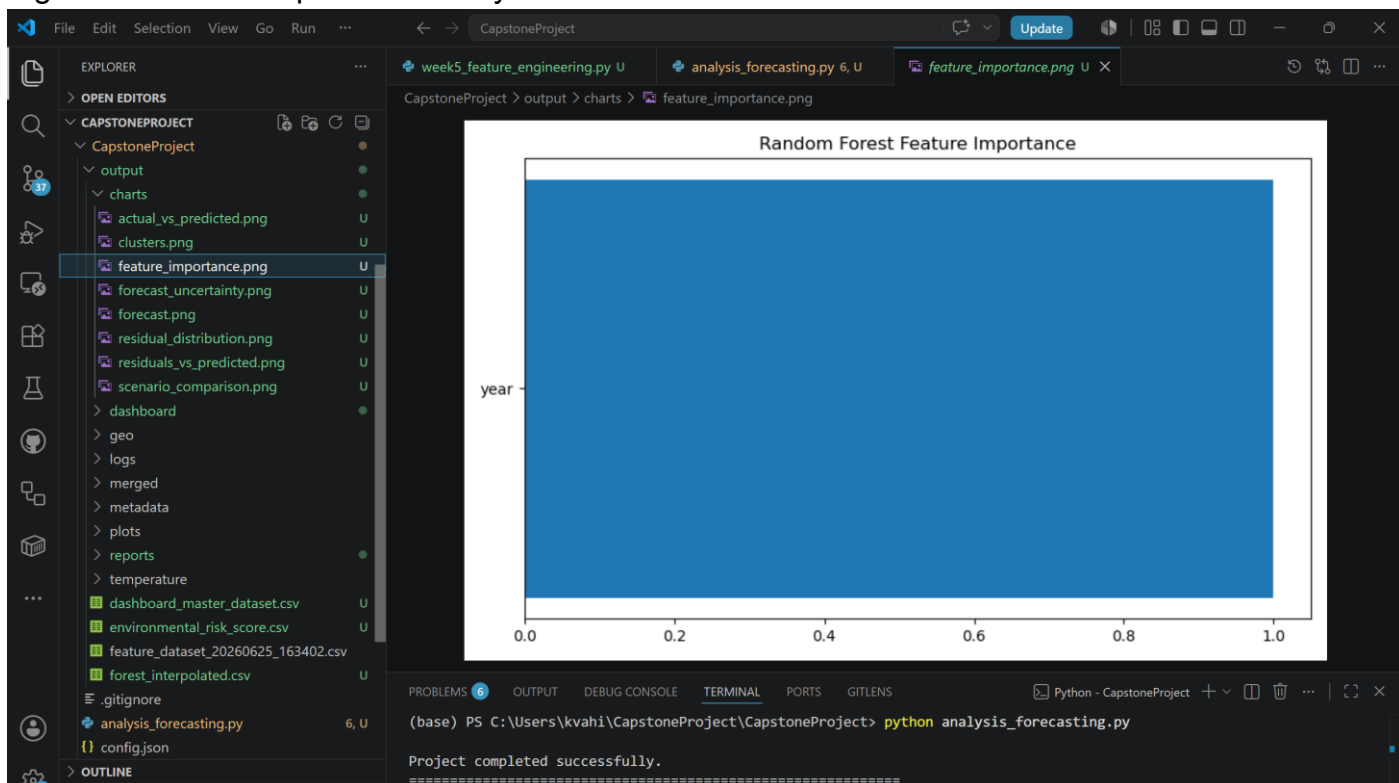
The residual distribution illustrates the forecasting errors produced by the model. Most residual values are concentrated around zero, indicating that prediction errors are minimal and that the forecasting model is stable and reliable.

Figure 5: Residuals vs Predicted



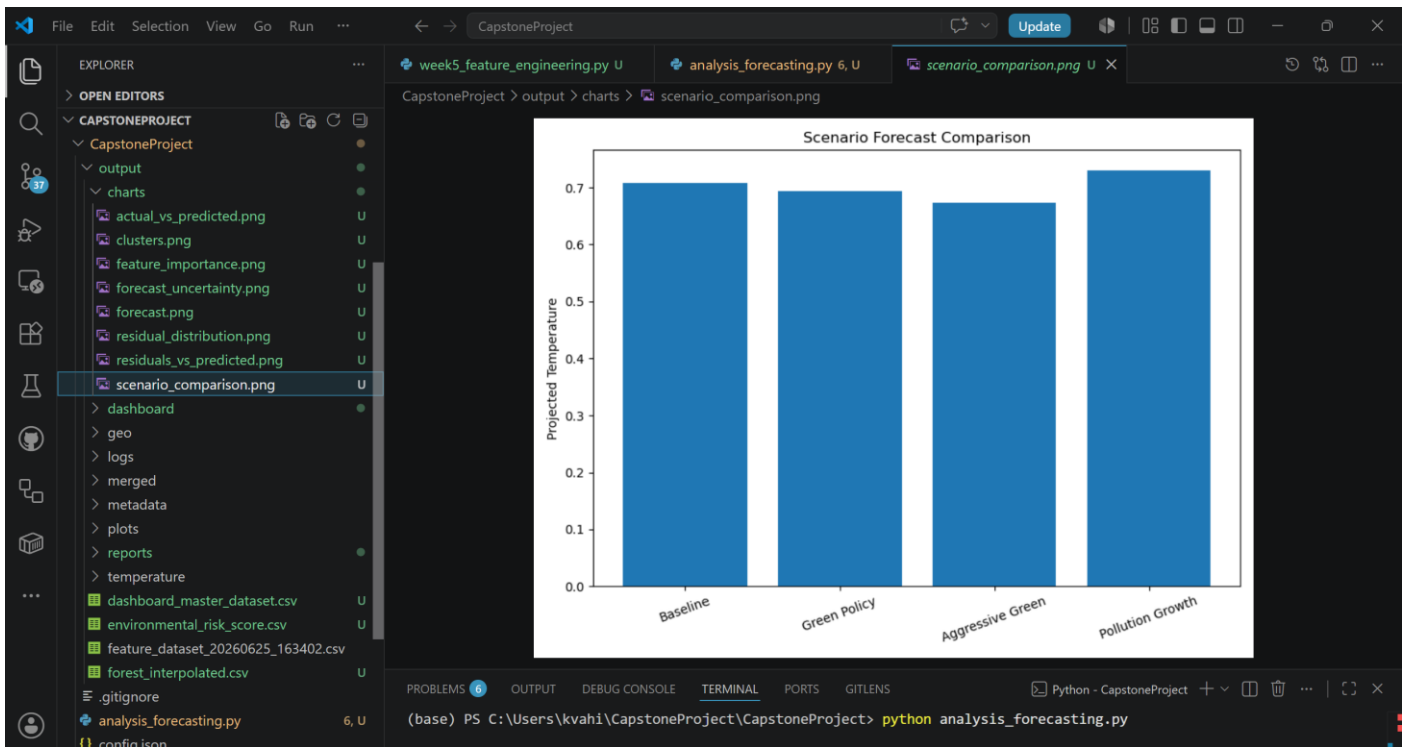
This diagnostic chart evaluates the relationship between residuals and predicted values. The absence of a strong pattern indicates that prediction errors are randomly distributed, suggesting that the model does not suffer from substantial bias.

Figure 6: Feature Importance Analysis



The feature importance chart illustrates the contribution of input variables to the Random Forest model. Since the current forecasting model uses the year variable as the primary predictor, it demonstrates the dominant influence of temporal progression on temperature anomalies.

Figure 7: Scenario Forecast Comparison

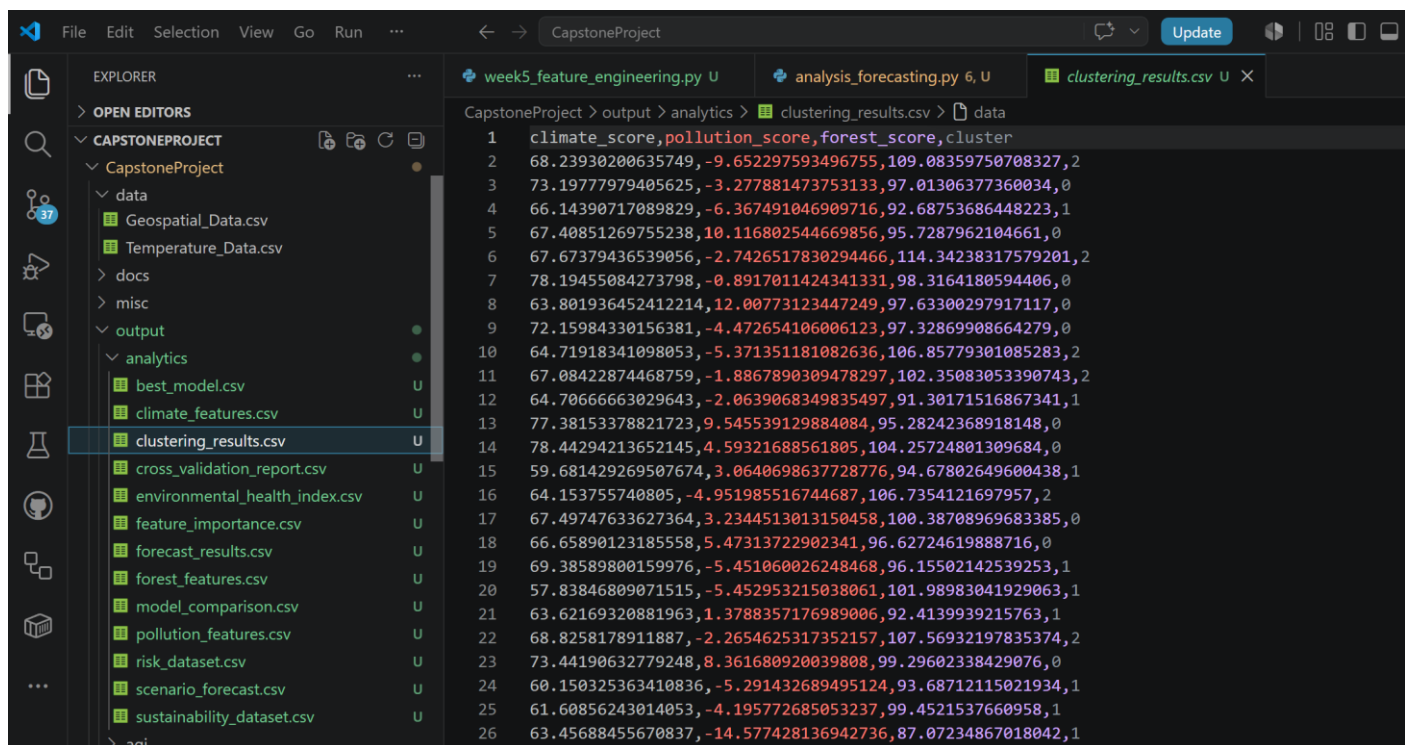
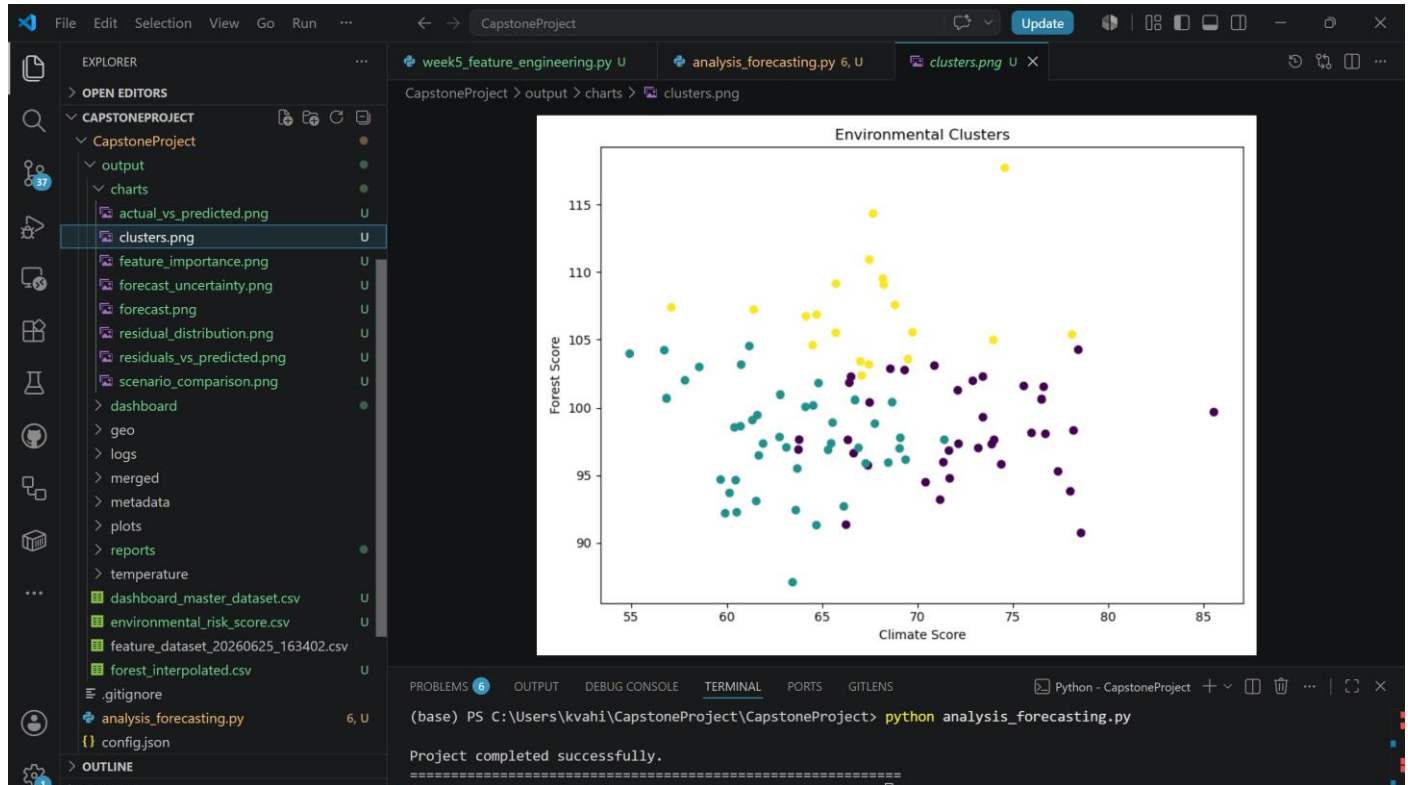


This visualization compares projected temperature outcomes across four environmental scenarios:

- Baseline
- Green Policy
- Aggressive Green
- Pollution Growth

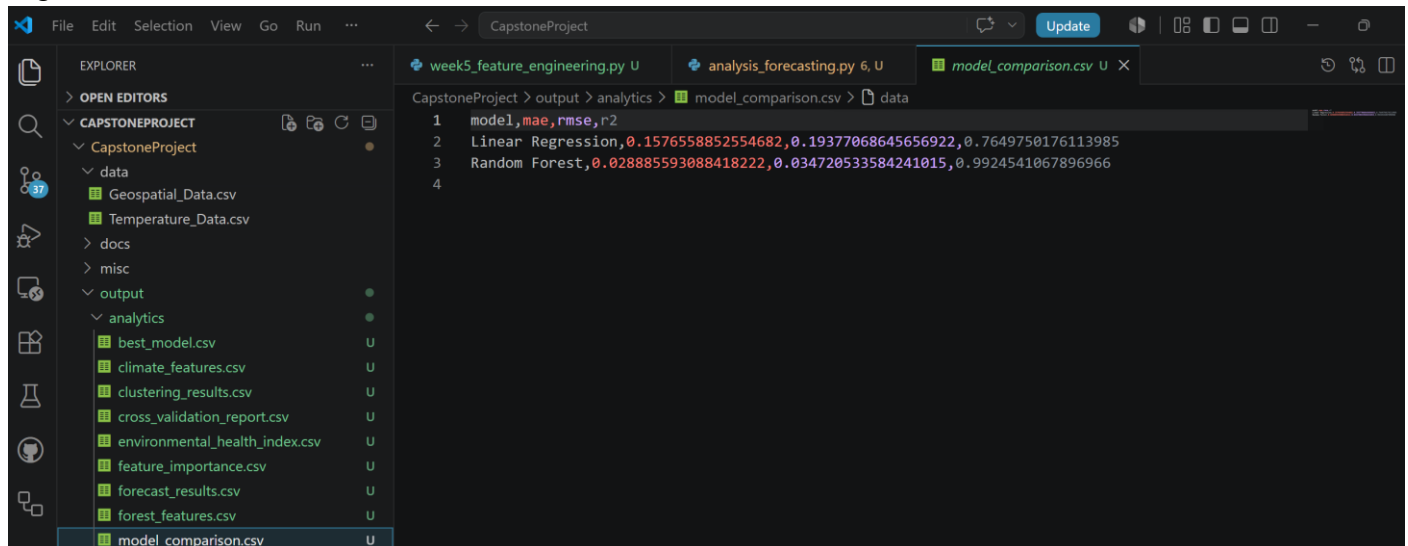
The Aggressive Green scenario produces the most favorable outcome, while Pollution Growth results in the highest projected environmental impact. This demonstrates the value of sustainability-focused interventions.

Figure 8: Environmental Cluster Analysis



K-Means clustering was applied to environmental indicators to identify distinct environmental segments. The chart shows three clusters representing different combinations of climate, pollution, and forest conditions. These clusters support environmental risk segmentation and targeted sustainability planning.

Figure 9: Model Evaluation Results



This table compares the performance of Linear Regression and Random Forest models using:

- Mean Absolute Error (MAE)
- Root Mean Square Error (RMSE)
- R-Squared (R^2)

Key result:

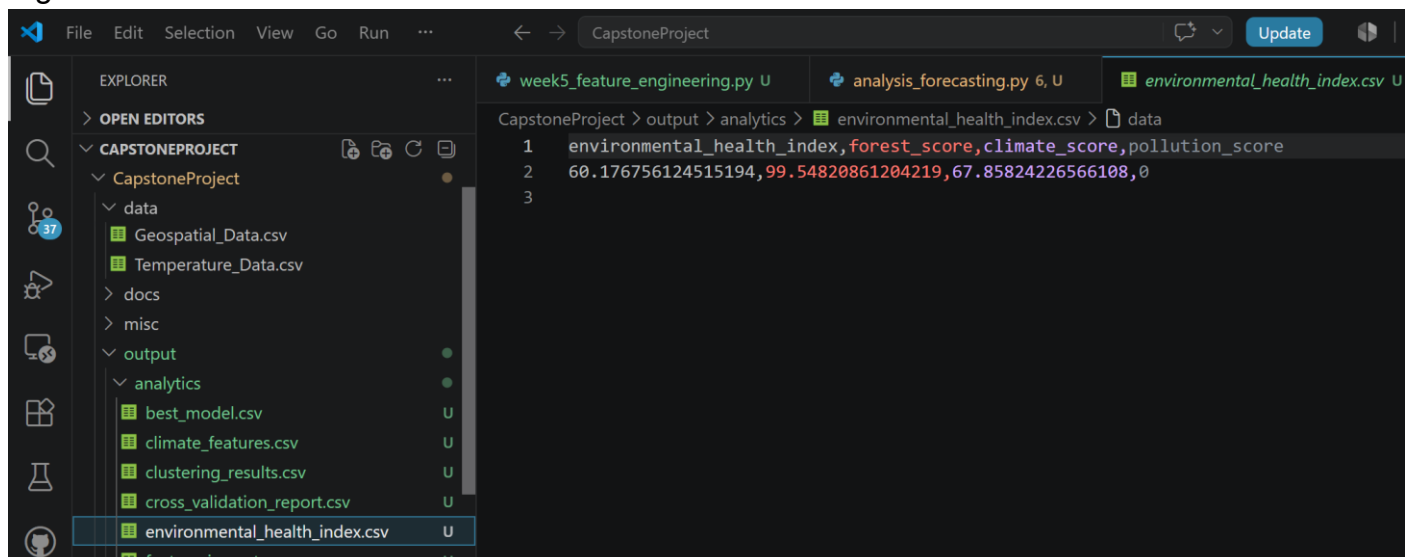
Best Model: Random Forest

$R^2 = 0.992454$

MAE = 0.028886

RMSE = 0.034721

Figure 10: Environmental Health Index

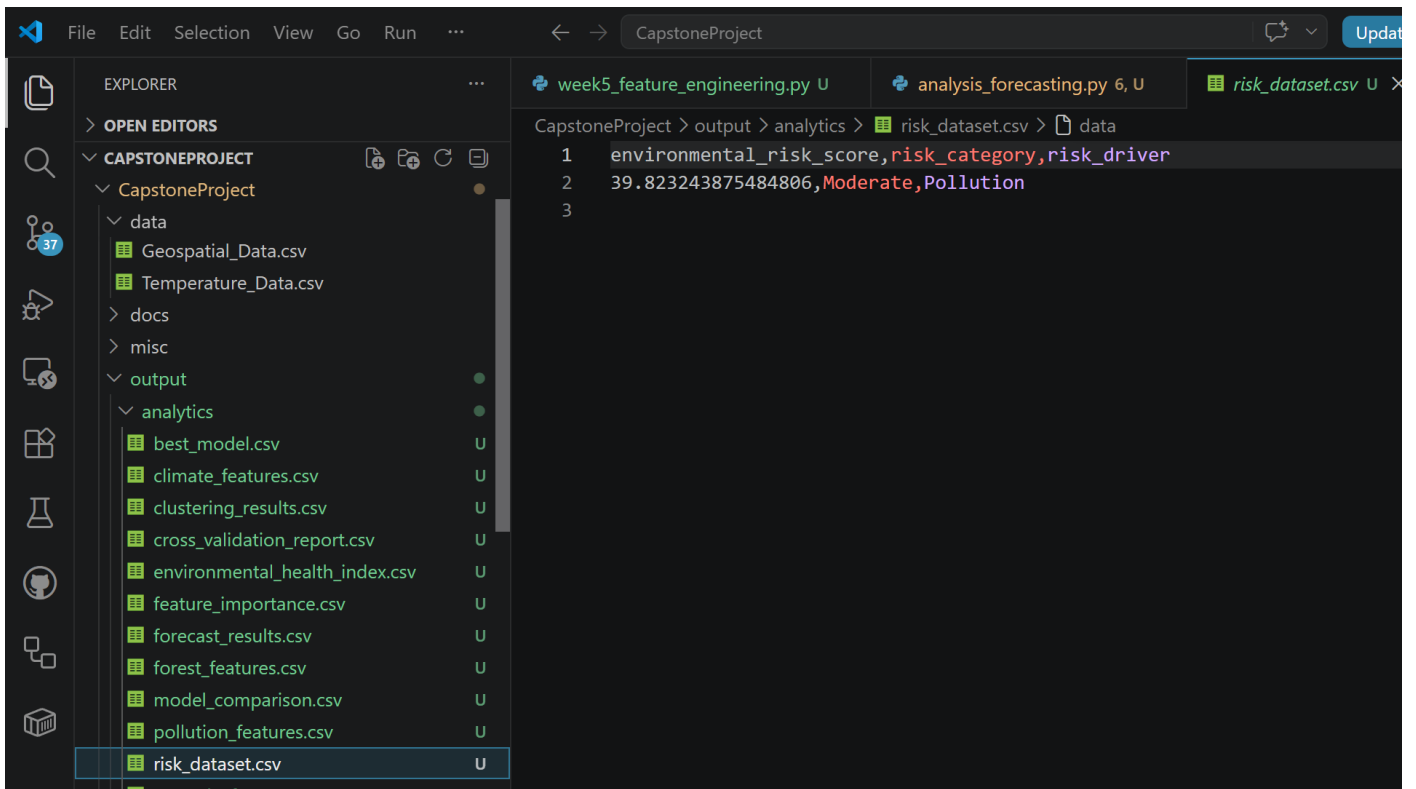


Provides the overall environmental health assessment generated from the Climate Severity Index, Pollution Exposure Index, and Forest Sustainability Index.

Result:

Environmental Health Index = 60.18

Figure 11: Risk Assessment Results

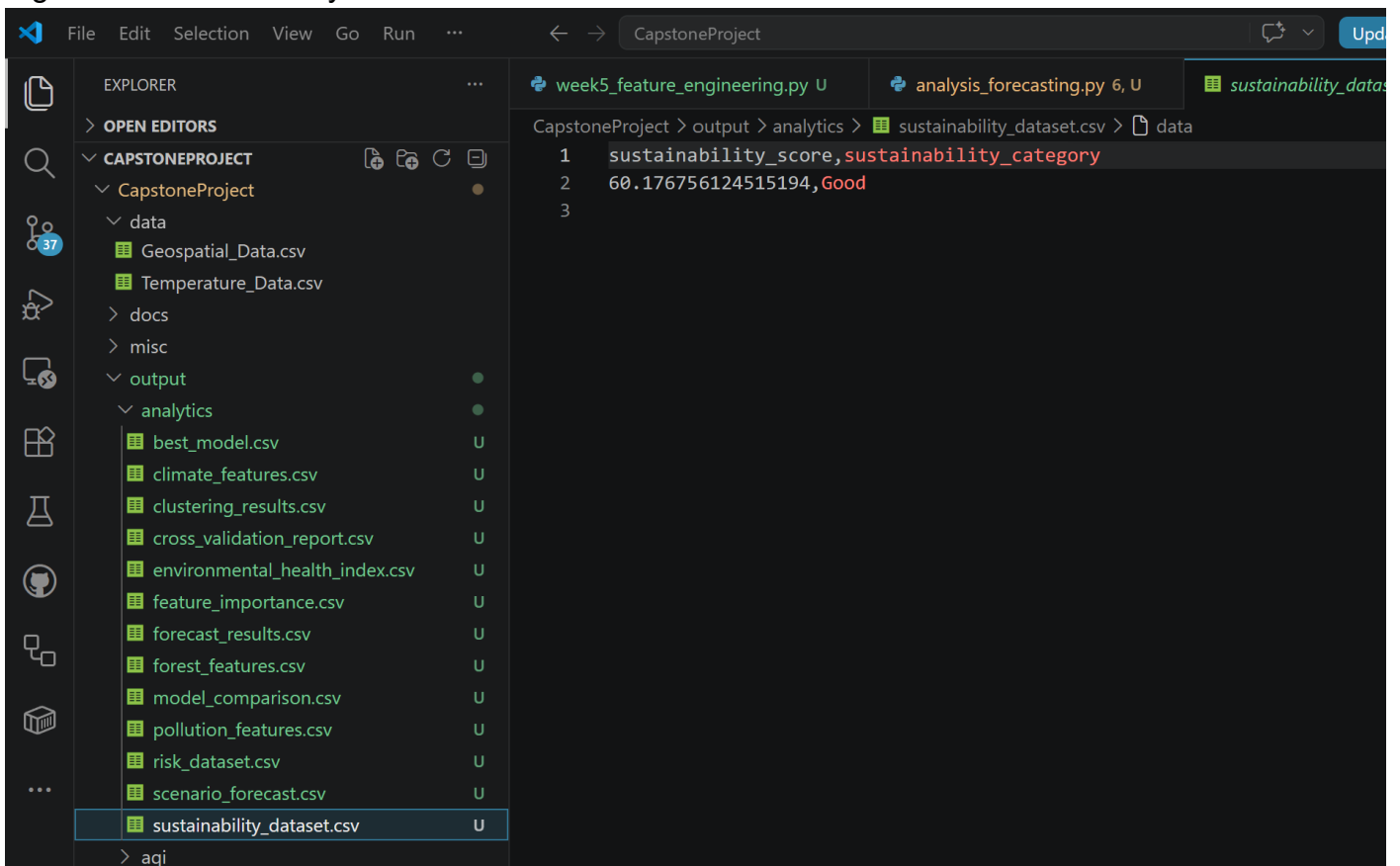


Contains the environmental risk assessment results.

Result:

Risk Category = Moderate

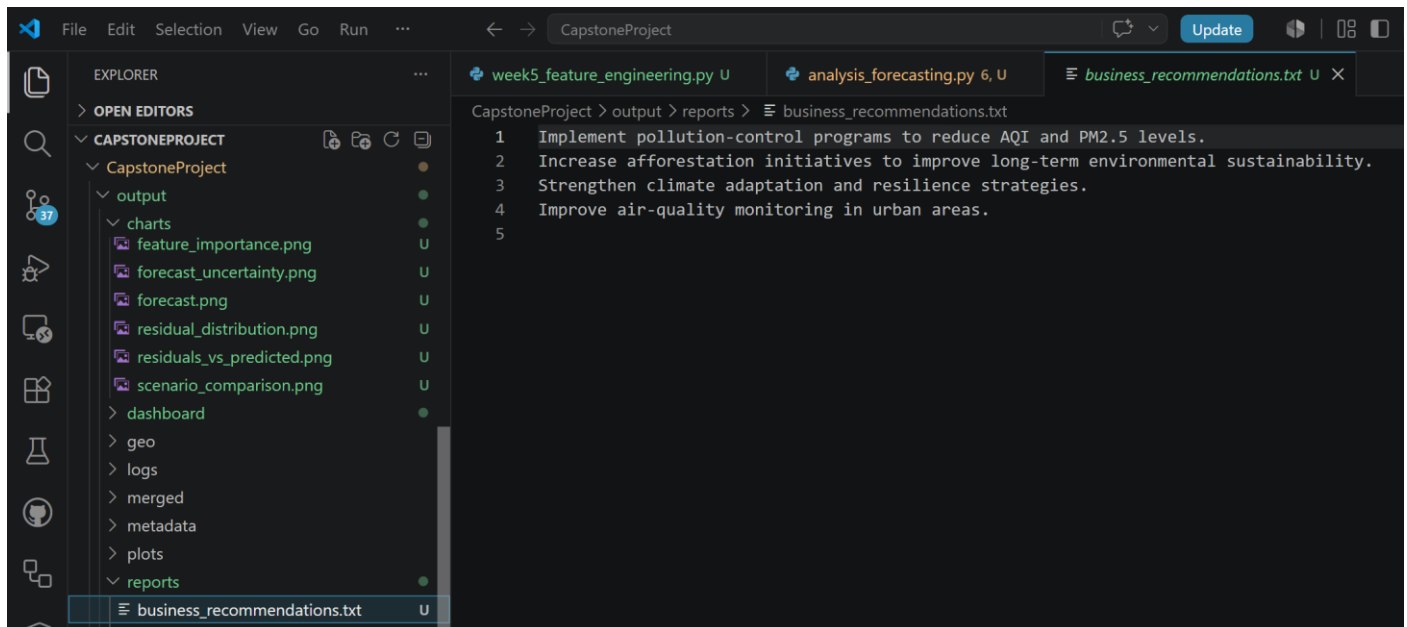
Figure 12: Sustainability Assessment Results



Displays overall sustainability performance.

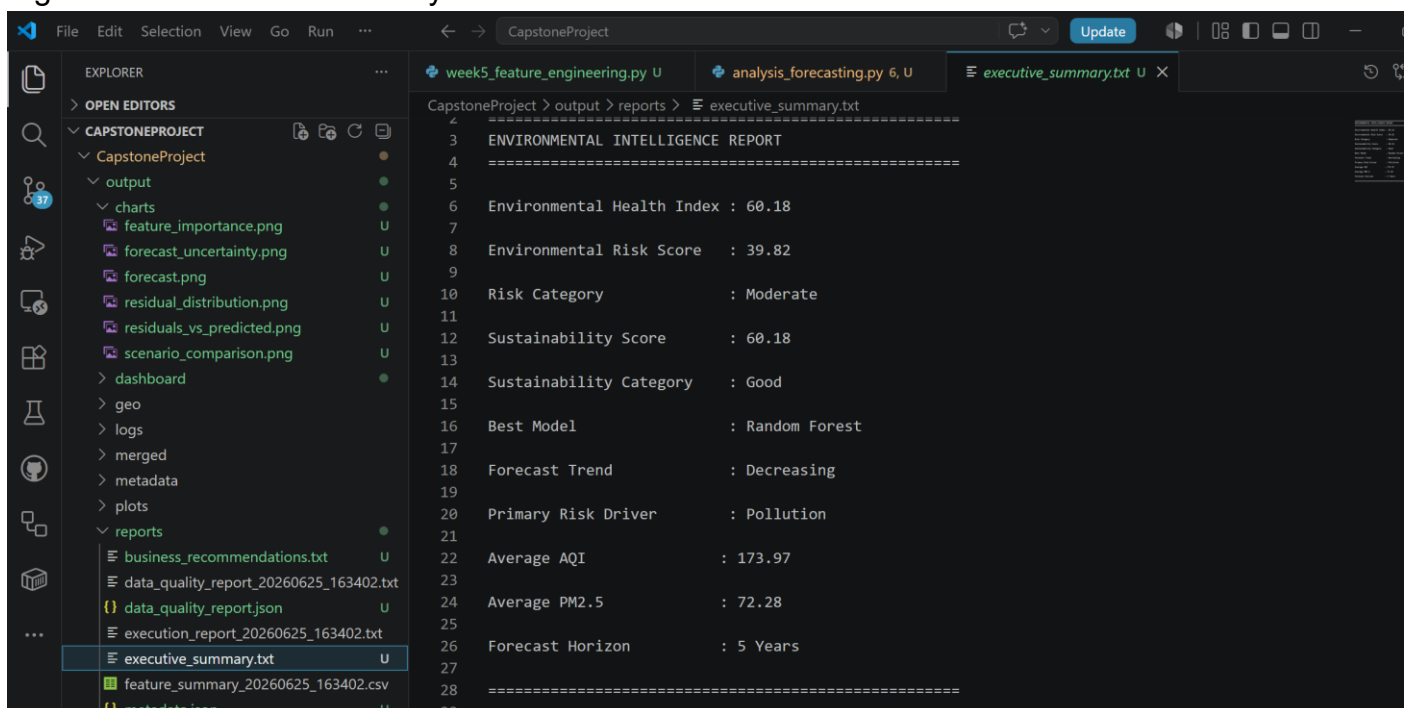
Result: Sustainability Category = Good

Figure 13: Business Recommendations



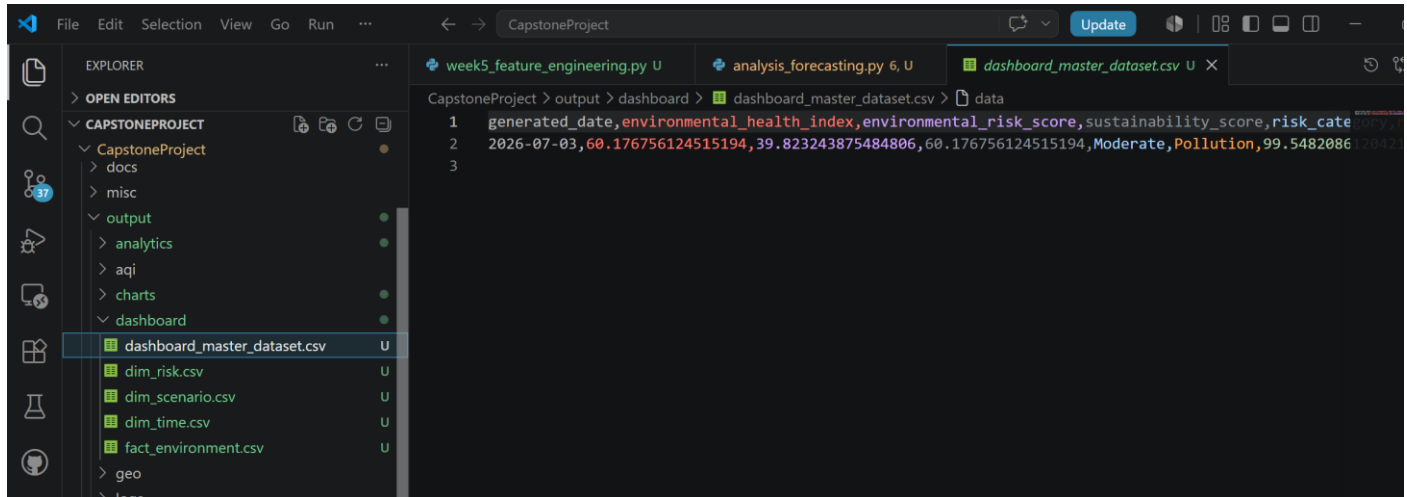
Automatically generated recommendations based on environmental risk analytics, pollution indicators, and sustainability metrics.

Figure 14: Executive Summary



Summarizes the major findings of the project including environmental health, risk levels, forecasting results, and strategic recommendations.

Figure 15: Power BI Data Model



The project generates a Power BI-ready star schema consisting of fact and dimension tables. These datasets will be imported into Power BI during Week 6 to create interactive sustainability and environmental intelligence dashboards.

The star schema supports:

- Executive KPI reporting
- Environmental risk monitoring
- Sustainability analysis
- Temperature forecasting
- Scenario analysis
- Environmental cluster visualization

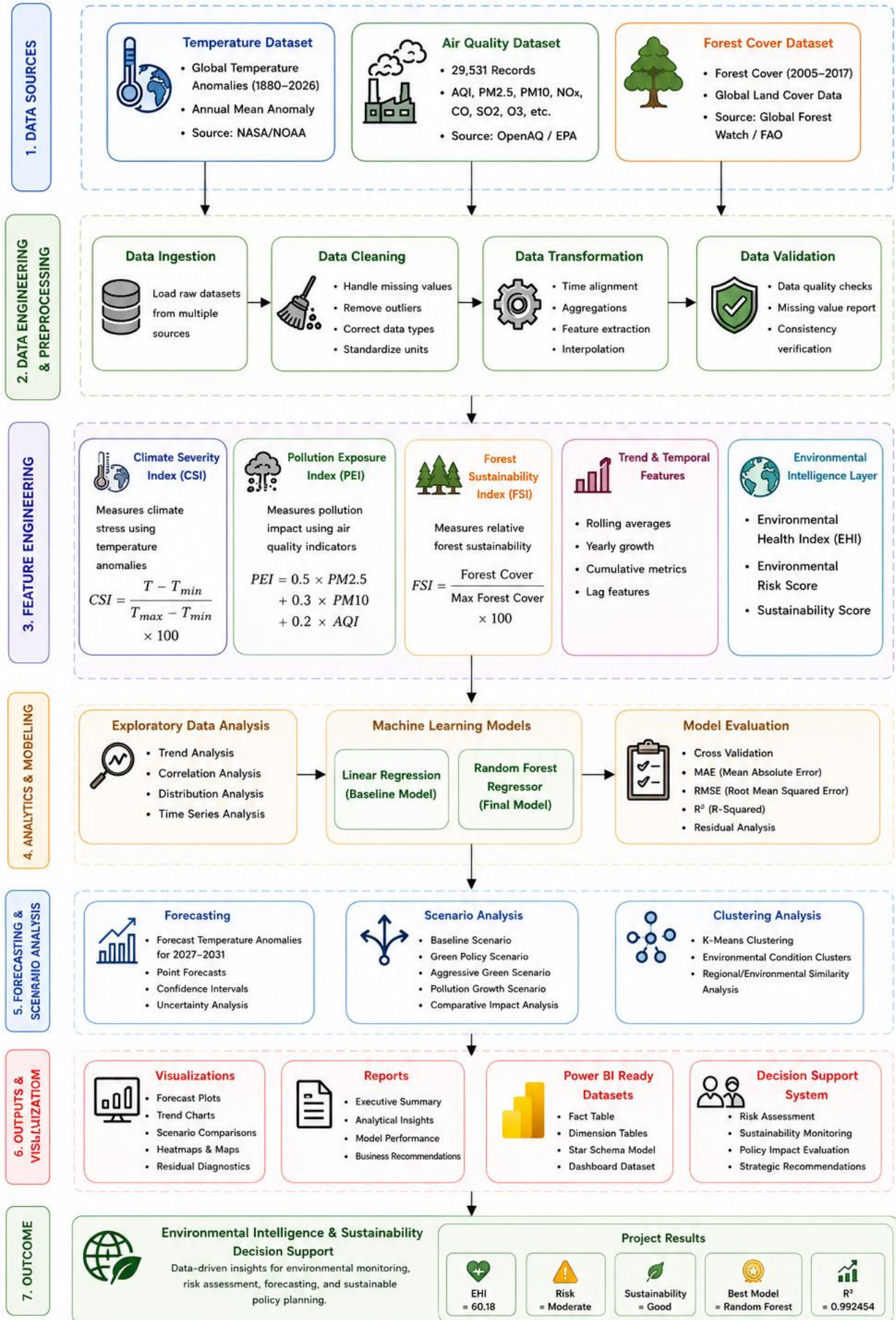
The data model provides a scalable foundation for business intelligence and decision support applications.

Week 5 Environmental Data Engineering, Analytics and Forecasting Pipeline

This image illustrates the complete Week 5 workflow implemented for the Environmental Data Analytics Capstone Project, covering data ingestion, preprocessing, feature engineering, machine learning, forecasting, scenario analysis, clustering, visualization, and decision-support outputs.

WEEK 5: DATA ENGINEERING, ANALYTICS & FORECASTING

Environmental Intelligence and Sustainability Analytics Platform



The figure illustrates the end-to-end analytical workflow developed during Week 5. Environmental datasets from multiple authoritative sources are first cleaned, validated, and transformed into a unified analytical dataset. Advanced feature engineering generates domain-specific environmental indicators that improve predictive capability. The engineered data is then used for exploratory analysis, machine learning model development, forecasting, scenario evaluation, and clustering. Model performance is assessed using cross-validation and standard evaluation metrics, while the final outputs include visualizations, reports, dashboard-ready datasets, and decision-support insights. This modular architecture demonstrates a production-oriented analytics pipeline that supports scalable environmental monitoring, forecasting, and sustainability